WHOLE HOUSE FAN

CALIFORNIA ENERGY COMMISSION

CEC-CF2R-MCH-31-H

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF INSTALLATION

Note: This table completed by ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Whole House Fan Measurement Procedures

01	Whole House Fan Airflow/Watts Measurement Procedure:	

B. Required Whole House Fan Specifications

01	02	03
Fan Name	WHF Modeled Airflow (CFM)	WHF Modeled Fan Power (Watts)

C1. Tested Whole House Fan Equipment Information – Airflow and watts measured per Whole House Fan

Requirements for Whole House Fans are given in Sections 150.1(b)2.B.vi and 150.1(c)12

01	02	03	04	05	06
Fan Name	Fan Location	WHF Manufacturer Name	WHF Model Number	WHF Tested Airflow (CFM) Per RA3.9.4.1	WHF Tested Watts Per RA3.9.4.2

C2. Tested Whole House Fan Equipment Information – Airflow measured per whole house fan and watts measured as a total value

Requirements for Whole House Fans are given in Sections 150.1(b)2.B.vi and 150.1(c)12

01	02	03	04	05	06
Fan Name	Fan Location	WHF Manufacturer Name	WHF Model Number	WHF Tested Airflow (CFM) Per RA3.9.4.1	WHF Tested Watts Per RA3.9.4.2

D. Whole House Fan Compliance Calculations

01	Required CFM	
02	Installed CFM	
03	Required Fan Efficacy (Watts/CFM)	
04	Installed Fan Efficacy (Watts/CFM)	

E. Compliance Statement

F. Additional Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The installed fan shall be listed in the Home Ventilating Institute Certified Products Directory
02	The homeowner shall be provided with user instructions documentation that describes the proper use of the whole house fan necessary to obtain the full energy savings benefit.

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this certificate of installation is true and correct.
2. I am either: a) a responsible person eligible under division 3 of the business and professions code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this certificate of installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person’s behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this certificate of installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the certificate of compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a ECC rater will check the installation to verify compliance and if such checking determines the installation fails to comply, I am required to offer any necessary corrective action at no charge to the building owner.
5. I understand that a registered copy of this certificate of installation shall be posted or made available with the building permit(s) issued for the building and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this certificate of installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone	Date Signed:
Third Party Quality Control Program (TPQCP) Status:	Name of TPQCP (if applicable):	

For assistance or questions regarding the Energy Standards, contact the Energy Hotline at: 1-800-772-3300

CF2R-MCH-31-H User Instructions

Section A. Whole House Fan Measurement Procedures

1. Select the procedure used to measure whole house fan Airflow.
2. Select the procedure used to measure whole house fan Watts.

Section B. Required Whole House Fan Specifications

1. Fan name will be auto populated from CF1R.
2. Whole House Fan (WHF) airflow in CFM will be auto populated from CF1R.
3. Whole House Fan (WHF) power in Watts will be auto populated from CF1R.

Section C1. Whole House Fan (WHF) Equipment Information (This section is required if procedure in A01 uses a Portable Meter)

1. Fan name will be auto populated from CF1R.
2. Enter the location where each whole house fan is installed in the house.
3. Enter the name of the manufacturer for each whole house fan installed in the house.
4. Enter the model number for each whole house fan installed in the house.
5. Enter the tested airflow in CFM per RA3.9.4.1 for each whole house fan installed in the house.
6. Enter the tested the Watts per RA3.9.4.2 for each whole house fan installed in the house.

Section C2. Whole House Fan (WHF) Equipment Information (This section is required if procedure in A01 uses a Revenue Meter)

1. Fan name will be auto populated from CF1R.
2. Enter the location where each whole house fan is installed in the house.
3. Enter the name of the manufacturer for each whole house fan installed in the house.
4. Enter the model number for each whole house fan installed in the house.
5. Enter the tested airflow in CFM per RA3.9.4.1 for each whole house fan installed in the house.
6. Enter the total tested Watts per RA3.9.4.2 for all whole house fans installed in the house.

Section D. Whole House Fan Compliance Calculations

1. This field is automatically populated from Section B.
2. This field is automatically populated from Section C.
3. This field is automatically populated from Section B.
4. This field is automatically calculated from section C.

E. Compliance Statement

To comply, the total installed whole house fan airflow must equal to or greater than the required airflow and the installed fan efficacy must be less than or equal to the required fan efficacy.

F. Additional Requirements

1. To qualify for the whole house fan compliance credit the installed whole house fan must be listed in Home Ventilating Institute Certified Products Directory, https://www.hvi.org/hvi-certified-products-directory/.
2. The homeowner shall be provided with user instructions documentation that describes the proper use of the whole house fan necessary to obtain the full energy savings benefit.
3. This must be a true statement to comply.

A. Whole House Fan Measurement Procedures		
01	Whole House Fan Airflow/Watts Measurement Procedure:	<<user selection, “Powered Flow Capture Hood and Portable Watt Meter”, “Traditional Flow Capture Hood and Portable Watt Meter” , “Powered Flow Capture Hood and Revenue Meter” or “Traditional Flow Capture Hood and Revenue Meter”, “Fan Flowmeter and Portable Watt Meter” , “Fan Flowmeter and Revenue Meter”>>

B. Required Whole House Fan Specifications		
01	02	03
Fan Name	WHF Modeled Airflow (CFM)	WHF Modeled Fan Power (Watts)
<<Auto populate from CF1R>>	<<Auto populate from CF1R>>	<<Auto populate from CF1R>>

C1. Tested Whole House Fan Equipment Information
Requirements for Whole House Fans are given in Sections 150.1(b)2.B.vi and 150.1(c)12
<<If A01 equals any of these values this section is required
Powered Flow Capture Hood and Portable Watt Meter, Traditional Flow Capture Hood and Portable Watt Meter, or Fan Flowmeter and Portable Watt Meter;>>

01	02	03	04	05	06
Fan Name	Fan Location	WHF Manufacturer Name	WHF Model Number	WHF Tested Airflow Per RA3.9.4.1 (CFM)	WHF Tested Watts Per RA3.9.4.2
<<Auto populate from CF1R>>	<<user input, Fan Location, string>>	<<user input, WHF Manufacturer Name, string>>	<<user input, WHF Model Number, string>>	<<user input, WHF Measured Airflow, numeric>>	<<user input, WHF Measured Watts, numeric>>

C2. Whole House Fan Equipment Information
Requirements for Whole House Fans are given in Sections 150.1(b)2.B.vi and 150.1(c)12
<<If A01 equals any of these values this section is required
Powered Flow Capture Hood and Revenue Meter, Traditional Flow Capture Hood and Revenue Meter, or Fan Flowmeter and Revenue Meter

01	02	03	04	05	06
Fan Name	Fan Location	WHF Manufacturer Name	WHF Model Number	WHF Tested Airflow (CFM) Per RA3.9.4.1	WHF Tested Watts Per RA3.9.4.2
<<Auto populate from CF1R>>	<<user input, Fan Location, string>>	<<user input, WHF Manufacturer Name, string>>	<<user input, WHF Model Number, string>>	<<user input, WHF Measured Airflow, numeric>>	<<user input, WHF Measured Watts, numeric>>

D. Whole House Fan Compliance Calculations		
01	Required CFM	<<calculated field: sum of all *WHF Modeled Airflow (CFM)* values from Table B >>
02	Installed CFM	<<calculated field: sum of all *WHF Tested Airflow (CFM)* values from Table C >>
03	Required Fan Efficacy (Watts/CFM)	<<calculated field: sum of all *WHF Modeled Fan Power (Watts)* / sum of all *WHF Modeled Airflow (CFM)* values from Table B >>
04	Installed Fan Efficacy (Watts/CFM)	<<calculated field: <<If A01 equals any of these values Powered Flow Capture Hood and Portable Watt Meter, Traditional Flow Capture Hood and Portable Watt Meter, or Fan Flowmeter and Portable Watt Meter value is sum of all *WHF Tested Watts* from Table C1 / *Installed CFM* from Table D Else If A01 equals any of these values

		Powered Flow Capture Hood and Revenue Meter, Traditional Flow Capture Hood and Revenue Meter, or Fan Flowmeter and Revenue Meter value is *WHF Tested Watts* from Table C2 / *Installed CFM* from Table D >>

E. Compliance Statement
<<calculated field: if *Installed CFM* ≥ *Required CFM* and *Required Fan Efficacy (Watts/CFM)* ≥ *Installed Fan Efficacy (Watts/CFM)*; then display result: “System passes whole house fan verification requirement.”; else display result: ”System fails whole house fan verification requirement.”>>

F. Additional Requirements	
01	The installed fan shall be listed in the Home Ventilating Institute Certified Products Directory.
02	The homeowner shall be provided with user instructions documentation that describes the proper use of the whole house fan necessary to obtain the full energy savings benefit.
The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.	